

Matrices Inversas

$\det \neq 0 \rightarrow$ no singular

$$A \cdot A^{-1} = A^{-1} \cdot A = I$$

Método Gauss-Jordan

$$\left(\begin{array}{ccc|ccc} a_{11} & a_{12} & a_{13} & 1 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & 0 & 1 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 & 1 \end{array} \right) \xrightarrow{\text{Original}} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & a_{11} & a_{12} & a_{13} \\ 0 & 1 & 0 & a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 & a_{31} & a_{32} & a_{33} \end{array} \right) \xrightarrow{\text{Inversa}}$$

Método para 2x2

$$A^{-1} = \frac{1}{\det A} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$$

Método de Adjuntas

$$A^{-1} = \frac{\text{adj}(A)}{\det A}$$

Propiedades

1) $\det(I) = 1$

2) $F_1 \leftrightarrow F_2 = (-1) \det$

3) $K \cdot F_1 = K \cdot \det$

4) $\det(KA) = K^n \cdot \det A$

5) $F_1 \rightarrow (0, 0, 0) \therefore \det = 0$

6) $F_1 = F_2 \therefore \det = 0$

7) $F_1 = K F_2 \therefore \det = 0$

8) $\det A^T = \det A$

9) $\det(AB) = \det A \cdot \det B$

Determinantes

3x3

2x2

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \quad |A| = a_{11} \cdot a_{12} - a_{12} \cdot a_{21}$$

$$(a \cdot e \cdot i + b \cdot f \cdot g + c \cdot d \cdot h) - (g \cdot e \cdot c + h \cdot f \cdot a + i \cdot d \cdot b)$$

n x n

$$\det A = \sum a_{ij} \cdot \overbrace{A_{ij}}^{\text{Cofactor}}$$

$$\det A = a_{11} \cdot A_{11} + a_{12} \cdot A_{12} + \dots + A_{ij} A_{ij}$$

Triangular

$$\begin{pmatrix} a & b & c \\ 0 & e & f \\ 0 & 0 & i \end{pmatrix} \quad \begin{pmatrix} a & 0 & 0 \\ d & e & 0 \\ g & h & i \end{pmatrix} \quad \det = a \cdot e \cdot i$$

Cofactores

$$(-1)^{i+j} \cdot M_{ij}$$

Adjuntas

①

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

②

$$A = \begin{pmatrix} \overbrace{A_{11}}^{\text{Cofactores}} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix}$$

③

$$\text{adj } A = \begin{pmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{pmatrix}$$